Notes on Monte Carlo Method for American Options

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Contents

1 Numerical Example **2**
1.1 Setting up the model 2
1.2 First inductive step 3
1.3 Second inductive step 4
1.4 Determining option price 5

2 Generalization of Monte Carlo Algorithm **5**
2.1 Generalized inductive step: 6
2.1.1 Defining conditional expectation function 6
2.1.2 Defining continuation and immediate exercise vectors . . 6
2.1.3 Completing inductive step 6

3 Generating Price paths **7**
3.1 Random shocks 7
3.2 Generating paths 7

4 Regression **8**
4.1 LSM Regression 8
4.2 The basis functions 11

1 Numerical Example

This section aims to explain the LSM Monte Carlo approach using a simple numerical example.

1.1 Setting up the model

Let us consider an American Put option with the following properties:

1. The underlying stock is non dividend paying
2. The put option is exercisable at times $t = 1, 2, 3$ where $t = 3$ is the expiration date of the option, $t = 0$ is the present moment, and each time step is one year
3. The yearly risk free rate is $r = 6\%$
4. The continuously compounded annual discount factor is $R = e^{-r} \approx 0.94176$
5. The strike price of the option is $K = 1.10$

For simplicity, we will demonstrate the algorithm using 8 generated price paths for the underlying stock. The simulated price paths are given in 8×4 matrix S, where rows represent paths and columns represent times $t = \{0, 1, 2, 3\}$

$$S = \begin{bmatrix} 1.00 & 1.09 & 1.08 & 1.34 \\ 1.00 & 1.16 & 1.26 & 1.54 \\ 1.00 & 1.22 & 1.07 & 1.03 \\ 1.00 & 0.93 & 0.97 & 0.92 \\ 1.00 & 1.11 & 1.56 & 1.52 \\ 1.00 & 0.76 & 0.77 & 0.90 \\ 1.00 & 0.92 & 0.84 & 1.01 \\ 1.00 & 0.88 & 1.22 & 1.34 \end{bmatrix}$$

Given the property that an option's value is equal to it's intrinsic value at expiration, we can find the cashflows from the option at $t = 3$. These cashflow values will be stored is 8×3 matrix C where rows represent paths and columns represent time steps.

$$C = \begin{bmatrix} c_{11} & c_{12} & 0.00 \\ c_{21} & c_{22} & 0.00 \\ c_{31} & c_{32} & 0.07 \\ c_{41} & c_{42} & 0.18 \\ c_{51} & c_{52} & 0.00 \\ c_{61} & c_{62} & 0.20 \\ c_{71} & c_{72} & 0.09 \\ c_{81} & c_{82} & 0.00 \end{bmatrix}$$

1.2 First inductive step

For this step we must determine for each path at $t = 2$ if it is optimal to exercise early or continue holding the option.

Let us start by defining two vectors. Let X be the vector of prices at $t = 2$. Let Y be a vector of the cash flows received at $t = 3$ if the option is not exercised at $t = 2$, discounted by the discount factor $R = 0.94176$.

$$X = \begin{bmatrix} 1.08 \\ \times \\ 1.07 \\ 0.97 \\ \times \\ 0.77 \\ 0.84 \\ \times \end{bmatrix} \qquad Y = \begin{bmatrix} 0.00 \\ \times \\ 0.07 \\ 0.18 \\ \times \\ 0.20 \\ 0.09 \\ \times \end{bmatrix} \cdot 0.94176 = \begin{bmatrix} 0.00 \\ \times \\ 0.0659 \\ 0.1695 \\ \times \\ 0.1883 \\ 0.0847 \\ \times \end{bmatrix}$$

The rows 2, 5, and 8 are crossed out. This is because it is neither optimal to exercise at $t = 2$ or $t = 3$. Because of this, these paths can be ignored from $t = 2$ onward.

To evaluate if it is optimal to continue, we must determine the value of $E[Y|X]$, or the expected value of Y given X. This can be done by least squares regression to find

$$E[Y|X] = -1.07 + 2.983X - 1.813X^2$$

The least squared regression method will be explained further in later chapters. Using our expected value function we can create the following vector E which represents the expected cash flows from continuance at $t = 2$. We can also create the vector I which represents the cash flows at $t = 2$ from exercise.

$$E = \begin{bmatrix} 0.0369 \\ \times \\ 0.0461 \\ 0.1176 \\ \times \\ 0.1520 \\ 0.1565 \\ \times \end{bmatrix} \qquad I = \begin{bmatrix} 0.02 \\ \times \\ 0.03 \\ 0.13 \\ \times \\ 0.33 \\ 0.26 \\ \times \end{bmatrix}$$

Using this we can determine if it is optimal to exercise or continue holding the option at $t = 2$ for each path. We can observe that it is optimal to exercise at $t = 2$ for paths 4, 6, and 7. While it is optimal

to hold at $t = 2$ for paths 1 and 3. Using this, we can substitute cashflow values at $t = 2$ into matrix C defined at the beginning of the problem.

$$C = \begin{bmatrix} c_{11} & 0.00 & 0.00 \\ c_{21} & 0.00 & 0.00 \\ c_{31} & 0.00 & 0.07 \\ c_{41} & 0.13 & 0.18 \\ c_{51} & 0.00 & 0.00 \\ c_{61} & 0.33 & 0.20 \\ c_{71} & 0.26 & 0.09 \\ c_{81} & 0.00 & 0.00 \end{bmatrix}$$

However, because an option can only be exercised once, we must zero out any cashflow values that come after already exercising. This is because the option can only be exercised once.

$$C = \begin{bmatrix} c_{11} & 0.00 & 0.00 \\ c_{21} & 0.00 & 0.00 \\ c_{31} & 0.00 & 0.07 \\ c_{41} & 0.13 & 0.00 \\ c_{51} & 0.00 & 0.00 \\ c_{61} & 0.33 & 0.00 \\ c_{71} & 0.26 & 0.00 \\ c_{81} & 0.00 & 0.00 \end{bmatrix}$$

1.3 Second inductive step

We will now repeat the inductive strategy to find cashflow values at $t = 1$. We will again define vector X as a vector of prices at $t = 1$ and Y as a vector of cash flows received at $t = 2$, discounted by factor $R = 0.94176$.

$$X = \begin{bmatrix} 1.09 \\ \times \\ \times \\ 0.93 \\ \times \\ 0.76 \\ 0.92 \\ 0.88 \end{bmatrix} \qquad Y = \begin{bmatrix} 0.00 \\ \times \\ \times \\ 0.13 \\ \times \\ 0.33 \\ 0.26 \\ 0.00 \end{bmatrix} \cdot 0.94176 = \begin{bmatrix} 0.00 \\ \times \\ \times \\ 0.122 \\ \times \\ 0.311 \\ 0.245 \\ 0.00 \end{bmatrix}$$

Again we have crossed out the rows where it is not optimal to exercise at $t = 1$ or $t = 2$

Again we will use least squares regression to find $E[Y|X]$

$$E[Y|X] = 2.038 - 3.335X + 1.356X^2$$

Then, we will again define vectors E and I. Vector E is the expected value of continuation at $t = 1$ computed using the expected value function defined before. Then the vector I represents possible cash flows at $t = 1$ from exercise.

$$E = \begin{bmatrix} 0.0139 \\ \times \\ \times \\ 0.1093 \\ \times \\ 0.2866 \\ 0.1175 \\ 0.1533 \end{bmatrix} \qquad I = \begin{bmatrix} 0.01 \\ \times \\ \times \\ 0.17 \\ \times \\ 0.34 \\ 0.18 \\ 0.22 \end{bmatrix}$$

From this we can see that it is optimal to exercise at $t = 1$ for paths 4, 6, 7, and 8 while it is optimal to continue holding on path 1. Using this we will fill in the remainder of our cashflow table.

$$C = \begin{bmatrix} 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.07 \\ 0.17 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 \\ 0.34 & 0.00 & 0.00 \\ 0.18 & 0.00 & 0.00 \\ 0.22 & 0.00 & 0.00 \end{bmatrix}$$

1.4 Determining option price

Now that the cashflow table is complete, we can simply discount each cashflow and back to $t = 0$. Then we can take the average, which gives the price of the option using LSM monte carlo.

2 Generalization of Monte Carlo Algorithm

We will write an algorithm for an American option with the following properties:

1. The underlying stock is non dividend paying
2. The option is exercisable at equidistant time steps $t_1 \leq t_2 \leq ... \leq t_N$
3. The discount factor is $R = e^{-r\Delta t}$
4. The strike price of the option is K
5. The Intrinsic[s] returns the intrinsic value of the option for price s. For Put options Intrinsic[s] $= K - s$ and vice versa for Call options

We will start by generating P price paths for the underlying stock

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1N} \\ s_{21} & s_{22} & \cdots & s_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ s_{P1} & s_{P2} & \cdots & s_{PN} \end{bmatrix}$$

Then we will generate a cashflow table, computing cash flows at terminal points and zeroing out remaining entries.

$$C = \begin{bmatrix} 0.00 & 0.00 & \cdots & \text{Intrinsic}(\, s_{1N}) \\ 0.00 & 0.00 & \cdots & \text{Intrinsic}(\, s_{2N}) \\ \vdots & \vdots & \ddots & \vdots \\ 0.00 & 0.00 & \cdots & \text{Intrinsic}(\, s_{PN}) \end{bmatrix}$$

2.1 Generalized inductive step:

This step is induction of the cashflow table to t_n

2.1.1 Defining conditional expectation function

Let

$$X = S[:, n] \qquad Y = C[:, n+1] \cdot R$$

For $p \in 1, 2, ..., P$, pop all values from X and Y where $Intrinsic(X[p]) = 0$
Let

$$\mathbb{E}[Y|X] = \min_{\alpha}(\alpha_0 + \alpha_1 X + \alpha_2 X^2 - Y)$$

2.1.2 Defining continuation and immediate exercise vectors

We can define $P \times 1$ vector E with following rule:

For each row p in E: If $C[p, n+1] \neq 0 \vee \text{Intrinsic}\,(S[p, n]) \geq 0$ Then $e_p = \mathbb{E}[Y|X = S(p, n)]$. Else $e_p = 0$

We can define $P \times 1$ vector I with following rule:

For each row p in I: $I[p] = \text{Intrinsic}\,(S[p, n])$

2.1.3 Completing inductive step

Using vectors I and E we can fill in column n of the cashflow matrix

Iterating through $p \in 1, 2, ..., P$

If $I[p] \geq E[p]$ then $C[p,n] = I[p]$

Then for iterate through each row of the cashflow table. Zero out all entries to the right of the first nonzero entry in each row.

3 Generating Price paths

The generation of price paths is done through discrete time geometric Brownian motion. This section serves as brief surface level notes of strategy to implement this for the Monte Carlo algorithm.

3.1 Random shocks

Brownian motion satisfies:

$$W_{t+\Delta t} - W_t = \sqrt{\Delta t} \cdot Z_t$$

Where Z represents random shocks approximated by the Gaussian distribution $\mathcal{N}(0,1)$.

$$\mathcal{N}(0,1) \iff \mathbb{E}[Z] = 0 \wedge \mathrm{Var}(Z) = 1$$

$\mathcal{N}(0,1)$ is just a random number with chosen with variance of 1 and no upwards or downwards bias. It can be used to approximate the Brownian motion equation:

$$W_{t+\Delta t} - W_t = \sqrt{\Delta t} \cdot \mathcal{N}(0,1)$$

$$Z_t \sim \mathcal{N}(0,1)$$

3.2 Generating paths

When generating price paths using GBM, we assume the stock price S_t follows:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Where: μ is drift, σ is volatility, and W_t is Brownian motion. From this we can simulate discrete time updates using the following equation:

$$S_{t+\Delta t} = S_t \exp\Big((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z_t\Big)$$

We can define risk neutral drift using $\mu = r - q$ where r is risk free rate and q is divided yield. Which makes the stock price motion into a submartingale satisfying:

$$\mathbb{E}[S_T] = S_0\, e^{\mu T}$$

Substituting in our approximations for μ and Z_t we get the following equation for stock price updates:

$$S_{t+\Delta t} = S_t \exp\Big((r - q - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\,\mathcal{N}(0,1)\Big)$$

4 Regression

4.1 LSM Regression

The regression portion will be done with the least squares method, hence the name of the pricing strategy. For generalization, we will consider the basis functions given by vector b.

$$b = \begin{bmatrix} b_0(x) & b_1(x) & b_2(x) & b_3(x) \end{bmatrix}$$

Vectors $X, Y \in \mathbb{R}^n$ are given. The goal is to find coefficients c_0, c_1, c_2, c_3 such that $||c_0 b_0(X) + c_1 b_1(X) + c_2 b_2(X) + c_3 b_3(X) - Y||$ is minimized. This problem can be written in matrix vector form as follows:

$$A = \begin{bmatrix} b_0(x_0) & b_1(x_0) & b_2(x_0) & b_3(x_0) \\ b_0(x_1) & b_1(x_1) & b_2(x_1) & b_3(x_1) \\ \vdots & \vdots & \vdots & \vdots \\ b_0(x_n) & b_1(x_n) & b_2(x_n) & b_3(x_n) \end{bmatrix} \quad c = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} \quad \Longrightarrow \quad \min_c ||Ac - y||$$

A closed form solution already exists in the form of a normal equation:

$$A^T A c = A^T y$$

To compute the left side:

$$A^T = \begin{bmatrix} b_0(x_0) & b_0(x_1) & \cdots & b_0(x_n) \\ b_1(x_0) & b_1(x_1) & \cdots & b_1(x_n) \\ b_2(x_0) & b_2(x_1) & \cdots & b_2(x_n) \\ b_3(x_0) & b_3(x_1) & \cdots & b_3(x_n) \end{bmatrix} A = \begin{bmatrix} b_0(x_0) & b_1(x_0) & b_2(x_0) & b_3(x_0) \\ b_0(x_1) & b_1(x_1) & b_2(x_1) & b_3(x_1) \\ \vdots & \vdots & \vdots & \vdots \\ b_0(x_n) & b_1(x_n) & b_2(x_n) & b_3(x_n) \end{bmatrix}$$

Let $B = A^T A$

$$B = \begin{bmatrix} A^T(1,:) \cdot A(:,1) & A^T(1,:) \cdot A(:,2) & A^T(1,:) \cdot A(:,3) & A^T(1,:) \cdot A(:,4) \\ A^T(2,:) \cdot A(:,1) & A^T(2,:) \cdot A(:,2) & A^T(2,:) \cdot A(:,3) & A^T(2,:) \cdot A(:,4) \\ A^T(3,:) \cdot A(:,1) & A^T(3,:) \cdot A(:,2) & A^T(3,:) \cdot A(:,3) & A^T(3,:) \cdot A(:,4) \\ A^T(4,:) \cdot A(:,1) & A^T(4,:) \cdot A(:,2) & A^T(4,:) \cdot A(:,3) & A^T(4,:) \cdot A(:,4) \end{bmatrix}$$

Then:

$$B_{11} = A^T(1,:) \cdot A(:,1) = \sum_{i=0}^{n} b_0(x_i)\, b_0(x_i)$$

$$B_{12} = B_{21} = A^T(1,:) \cdot A(:,2) = \sum_{i=0}^{n} b_0(x_i)\, b_1(x_i)$$

$$B_{13} = B_{31} = A^T(1,:) \cdot A(:,3) = \sum_{i=0}^{n} b_0(x_i)\, b_2(x_i)$$

$$B_{14} = B_{41} = A^T(1,:) \cdot A(:,4) = \sum_{i=0}^{n} b_0(x_i)\, b_3(x_i)$$

$$B_{22} = A^T(2,:) \cdot A(:,2) = \sum_{i=0}^{n} b_1(x_i)\, b_1(x_i)$$

$$B_{23} = B_{32} = A^T(2,:) \cdot A(:,3) = \sum_{i=0}^{n} b_1(x_i)\, b_2(x_i)$$

$$B_{24} = B_{42} = A^T(2,:) \cdot A(:,4) = \sum_{i=0}^{n} b_1(x_i)\, b_3(x_i)$$

$$B_{33} = A^T(3,:) \cdot A(:,3) = \sum_{i=0}^{n} b_2(x_i)\, b_2(x_i)$$

$$B_{34} = B_{43} = A^T(3,:) \cdot A(:,4) = \sum_{i=0}^{n} b_2(x_i)\, b_3(x_i)$$

$$B_{44} = A^T(4,:) \cdot A(:,4) = \sum_{i=0}^{n} b_3(x_i)\, b_3(x_i)$$

Which allows us to define $A^T A$:

$$A^T A = \begin{bmatrix} \sum_{i=0}^{n} b_0(x_i)^2 & \sum_{i=0}^{n} b_0(x_i) b_1(x_i) & \sum_{i=0}^{n} b_0(x_i) b_2(x_i) & \sum_{i=0}^{n} b_0(x_i) b_3(x_i) \\ \sum_{i=0}^{n} b_1(x_i) b_0(x_i) & \sum_{i=0}^{n} b_1(x_i)^2 & \sum_{i=0}^{n} b_1(x_i) b_2(x_i) & \sum_{i=0}^{n} b_1(x_i) b_3(x_i) \\ \sum_{i=0}^{n} b_2(x_i) b_0(x_i) & \sum_{i=0}^{n} b_2(x_i) b_1(x_i) & \sum_{i=0}^{n} b_2(x_i)^2 & \sum_{i=0}^{n} b_2(x_i) b_3(x_i) \\ \sum_{i=0}^{n} b_3(x_i) b_0(x_i) & \sum_{i=0}^{n} b_3(x_i) b_1(x_i) & \sum_{i=0}^{n} b_3(x_i) b_2(x_i) & \sum_{i=0}^{n} b_3(x_i)^2 \end{bmatrix}$$

Which then allows us to define $A^T Ac$ as:

$$\begin{bmatrix} c_0 \sum_{i=0}^{n} b_0(x_i)^2 + c_1 \sum_{i=0}^{n} b_0(x_i)b_1(x_i) + c_2 \sum_{i=0}^{n} b_0(x_i)b_2(x_i) + c_3 \sum_{i=0}^{n} b_0(x_i)b_3(x_i) \\ c_0 \sum_{i=0}^{n} b_1(x_i)b_0(x_i) + c_1 \sum_{i=0}^{n} b_1(x_i)^2 + c_2 \sum_{i=0}^{n} b_1(x_i)b_2(x_i) + c_3 \sum_{i=0}^{n} b_1(x_i)b_3(x_i) \\ c_0 \sum_{i=0}^{n} b_2(x_i)b_0(x_i) + c_1 \sum_{i=0}^{n} b_2(x_i)b_1(x_i) + c_2 \sum_{i=0}^{n} b_2(x_i)^2 + c_3 \sum_{i=0}^{n} b_2(x_i)b_3(x_i) \\ c_0 \sum_{i=0}^{n} b_3(x_i)b_0(x_i) + c_1 \sum_{i=0}^{n} b_3(x_i)b_1(x_i) + c_2 \sum_{i=0}^{n} b_3(x_i)b_2(x_i) + c_3 \sum_{i=0}^{n} b_3(x_i)^2 \end{bmatrix}$$

Then we can define the right side of the equation:

$$A^T y = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \end{bmatrix}$$

$$S_m = \sum_{i=0}^{n} b_m(x_i) y_i$$

So to solve for c_0, c_1, c_2, c_3, solve:

$$\begin{bmatrix} c_0 \sum_{i=0}^{n} b_0(x_i)^2 + c_1 \sum_{i=0}^{n} b_0(x_i)b_1(x_i) + c_2 \sum_{i=0}^{n} b_0(x_i)b_2(x_i) + c_3 \sum_{i=0}^{n} b_0(x_i)b_3(x_i) \\ c_0 \sum_{i=0}^{n} b_1(x_i)b_0(x_i) + c_1 \sum_{i=0}^{n} b_1(x_i)^2 + c_2 \sum_{i=0}^{n} b_1(x_i)b_2(x_i) + c_3 \sum_{i=0}^{n} b_1(x_i)b_3(x_i) \\ c_0 \sum_{i=0}^{n} b_2(x_i)b_0(x_i) + c_1 \sum_{i=0}^{n} b_2(x_i)b_1(x_i) + c_2 \sum_{i=0}^{n} b_2(x_i)^2 + c_3 \sum_{i=0}^{n} b_2(x_i)b_3(x_i) \\ c_0 \sum_{i=0}^{n} b_3(x_i)b_0(x_i) + c_1 \sum_{i=0}^{n} b_3(x_i)b_1(x_i) + c_2 \sum_{i=0}^{n} b_3(x_i)b_2(x_i) + c_3 \sum_{i=0}^{n} b_3(x_i)^2 \end{bmatrix} = \begin{bmatrix} \sum_{i=0}^{n} b_0(x_i)y_i \\ \sum_{i=0}^{n} b_1(x_i)y_i \\ \sum_{i=0}^{n} b_2(x_i)y_i \\ \sum_{i=0}^{n} b_3(x_i)y_i \end{bmatrix}$$

So to solve the regression c_0, c_1, c_2, c_3 must be solved from the following system of equations:

$$c_0 B_{11} + c_1 B_{12} + c_2 B_{13} + c_3 B_{14} = \sum_{i=0}^{n} b_0(x_i) y_i$$

$$c_0 B_{21} + c_1 B_{22} + c_2 B_{23} + c_3 B_{24} = \sum_{i=0}^{n} b_1(x_i) y_i$$

$$c_0 B_{31} + c_1 B_{32} + c_2 B_{33} + c_3 B_{34} = \sum_{i=0}^{n} b_2(x_i) y_i$$

$$c_0 B_{41} + c_1 B_{42} + c_2 B_{43} + c_3 B_{44} = \sum_{i=0}^{n} b_3(x_i) y_i$$

Which can also be retpresented as:

Hence the system of equations gives the necessary coefficients needed to perform regression in the inductive step for a general set of 4 basis functions.

4.2 The basis functions

For the implementation of the LSM Monte Carlo algorithm, I have implemented the option to select one of the following basis functions

1. Quadratic
2. Cubic
3. Laguerre degree 2
4. Laguerre degree 3
5. Hermite degree 2
6. Hermite degree 3
7. Legendre degree 2
8. Legendre degree 3

Quadratic

$$b = \begin{bmatrix} 1 & x & x^2 & 0 \end{bmatrix}$$

Cubic

$$b = \begin{bmatrix} 1 & x & x^2 & x^3 \end{bmatrix}$$

Laguerre degree 2

$$b = \begin{bmatrix} \exp\left(\frac{-x}{2}\right) & \exp\left(\frac{-x}{2}\right)(1-x) & \exp\left(\frac{-x}{2}\right)\left(1-2x+\frac{x^2}{2}\right) & 0 \end{bmatrix}$$

Laguerre degree 3

$$b = \begin{bmatrix} \exp\left(\frac{-x}{2}\right) & \exp\left(\frac{-x}{2}\right)(1-x) & \exp\left(\frac{-x}{2}\right)\left(1-2x+\frac{x^2}{2}\right) & \exp\left(\frac{-x}{2}\right)\left(1-3x+\frac{3x^2}{2}-\frac{x^3}{6}\right) \end{bmatrix}$$

Hermite degree 2

$$b = \begin{bmatrix} 1 & x & x^2-1 & 0 \end{bmatrix}$$

Hermite degree 3

$$b = \begin{bmatrix} 1 & x & x^2-1 & x^3-3x \end{bmatrix}$$

Legendre degree 2

$$b = \begin{bmatrix} 1 & x & \frac{3x^2-1}{2} & 0 \end{bmatrix}$$

Legendre degree 3

$$b = \begin{bmatrix} 1 & x & \frac{3x^2-1}{2} & \frac{5x^3-3x}{2} \end{bmatrix}$$